CABRERA, JEN JADE B.

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EDUCATION

San Pablo Colleges · Magna Cum Laude — Hermanos Belen Street, San Pablo City
Bachelor of Science in Computer Science — September 2022 - June 2025

WORK EXPERIENCE

Senior Associate Solutions Architect — PDAX Inc, January 2026 - Present

- **Reduced Mean Time to Remediation (MTTR) by 80% by architecting and engineering a serverless, event-driven** security ecosystem using **Hexagonal Architecture** with **multiple entry points (CLI and AWS EventBridge)**; utilized a fan-out pattern with **Lambda** and **SQS** to automate the ingestion, routing and **idempotency** (**DynamoDB**), and near real-time auto-resolution of **thousands of monthly findings** via **Jira.**

Senior Associate Software Quality Engineer — PDAX Inc, March 2025 - December 2025

- **Architected the organization's foundational QE platform** by integrating **Docker, Nginx, Jenkins, Vault, Grafana, Grafana Loki, and InfluxDB** to centralize test execution and observability; reduced environment setup time by **70%** and unified reporting across **6+ cross-functional squads**.
- **Designed and built a secure, standardized automation ecosystem** for backend and mobile (Robot Framework, Appium, BrowserStack) within a **complex monorepo**; utilized **Hashicorp Vault** to automate 100% of secret injection, eliminating manual credential exposure across all pipelines.
- **Engineered an on-demand, high-concurrency performance stack** using **Grafana K6** to identify critical system bottlenecks by simulating **thousands of concurrent users**, significantly increasing system reliability and stability during peak traffic periods.

Software Quality Engineer Intern — PDAX Inc, February - March 2025

- **Developed a hybrid CI/CD pipeline PoC** using **Jenkins, Docker, and SonarQube** to automate smoke tests on PRs and full regression suites for Staging/Production deployments.
- **Engineered and architected a multi-tier CI gate system** for test repositories, enforcing QA-led test selection and approval workflows for feature, develop, and master branches.
- **Built a monorepo PoC** using **Python (UV) and Robot Framework Requests**; integrated **HashiCorp Vault** for centralized secrets and modular coding abstractions.

Full Stack & Machine Learning Engineer — Freelance, September - December 2024

- **Developed an end-to-end Legal Document Evaluation System** using **React, Flask-RESTX, and Docker**, automating the assessment of project proposals with a custom NLP pipeline.
- **Engineered a retrieval pipeline** utilizing **GloVe embeddings** for tokenization and **SBERT (Sentence-BERT)** for deep semantic similarity to match complex legal clauses with project requirements.
- **Deployed a secure, containerized infrastructure** using **Docker and Cloudflare Tunnels**, enabling encrypted, firewall-friendly remote access for the evaluation system.
- **Conducted Exploratory Data Analysis (EDA)** using **Jupyter, Pandas, and Seaborn** to refine the NLP model, reducing false-positive document classifications by **40%** during the prototyping phase.

INTERESTS AND TECHNOLOGIES

Interests: Data Science, Full Stack Engineering, Quality Engineering, DevSecOps, Architecture, Design Patterns
Languages: Python, Javascript, TypeScript, PHP, Bash, SQL, Java, R, Dart, C#, Arduino
Cloud & Infrastructure: AWS Ecosystem, Docker, Jenkins, HashiCorp Vault, N8N, Grafana, Loki, Git
Frameworks: React, Next.js, Nest.js, Flask, FastAPI, Laravel, Flutter, JavaFX, TailwindCSS, K6, Robot Framework
Databases: PostgreSQL, MySQL, NoSQL, InfluxDB, Supabase, PocketBase,
Data Science & AI: Pandas, Seaborn, Scikit-learn, Streamlit, Jupyter, PydanticAI, Ggplot, Plotly
Systems & Tools: Linux (Arch/CachyOS/Ubuntu/Zorin), Tmux, Neovim, Adobe Ecosystem

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PROMINENT PROJECTS

Anonalyze: An AI and NLP-Enhanced Platform for Sentiment and Insight Extraction 2024

- **Project repository: https://github.com/hyoaru/anonalyze**
- Designed and implemented a **supervised machine learning model** using **Pandas**, **NumPy**, **Scikit-Learn**, and **Jupyter Notebook** with a **Multinomial Naive Bayes** classifier. The model predicts the **sentiment** and **emotion** of sentences using an emotion dataset from Kaggle.
- Developed an **API** for the model using **Flask**, **Flask-RESTX**, **Gunicorn**, **Docker**, and documented it using **OpenAPI/Swagger**.
- Built the core **backend API** using **PHP Laravel** and **OpenAPI/Swagger** for robust data handling and scalability.
- Developed a **web client** using **TypeScript**, **React**, **TanStack Router**, **TanStack Query**, **TailwindCSS**, and **ShadcnUI** for a dynamic, responsive user interface.
- Designed the platform to enable **executives** to post questions and allow their **subjects** to respond. The platform predicts the **sentiment** and **emotion** of the responses, extracts **key concepts**, and generates **keywords** and **keyphrases**.
- Incorporated a **summarization feature** using a **Large Language Model (LLM)** from OpenAI to generate summaries of the entire thread.

Beyond Decor: A Portfolio and Inquiry Website System 2023

- **Project repository: https://github.com/hyoaru/beyond-decor**
- Developed a **portfolio and inquiry website system** for **Beyond Decor**, a party and entertainment service, using **ReactJS**, **Next.js**, **DaisyUI**, **TailwindCSS**, and **PocketBase**.
- This project served as an eye-opener to the **composability design principle** in **React** and deepened my understanding of **Next.js** philosophies for building optimized, scalable web applications.
- The website showcases **Beyond Decor's services**, allowing users to explore party and entertainment options, inquire about services, and get in touch with the company.

Philippine Poverty Area Estimates Choropleth 2023

- **Project repository: https://github.com/hyoaru/philippine-poverty-area-estimates-choropleth**
- Developed a **web application** providing a visual representation of the estimated magnitude of poor families in the Philippines using a **choropleth map**.
- The map visualizes data from the years **2006**, **2009**, **2012**, and **2015** to give users insights into the poverty distribution across regions.
- **Data source:** United Nations Office for the Coordination of Humanitarian Affairs (**UN OHCA**) and **Philippine Statistics Authority (PSA)**.
- The project was built using **Python**, **Jupyter Notebook**, **NumPy**, **Pandas**, and **Streamlit**.

Breast Cancer Classification: Supervised Machine Learning 2022

- **Project repository: https://github.com/hyoaru/sparta-supervisedml-binary-classification**
- Completed a **peer-reviewed machine learning task** as part of the **Smarter Philippines through Data Analytics R&D, Training and Adoption (SPARTA)** program on the course **Data Science and Machine Learning with Python**.
- Implemented **binary classification** using the **Breast Cancer Wisconsin Diagnostic Dataset**, employing machine learning techniques to predict the presence of cancer based on feature data.
- This project sparked my interest in **machine learning** and helped me discover my passion for the field. As a **first-year scholar** in the SPARTA program, I had the opportunity to collaborate with peers who were already working professionals, which enriched my learning experience and broadened my perspective.

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CERTIFICATIONS

Google Data Analytics Capstone: Complete a Case Study	July 2023 · Google
Data Analysis with R Programming	June 2023 · Google
Share Data Through the Art of Visualization	May 2023 · Google
Analyze Data to Answer Questions	April 2023 · Google
Process Data from Dirty to Clean	March 2023 · Google
Prepare Data for Exploration	January 2023 · Google
Foundations: Data, Data, Everywhere	December 2022 · Google
Ask Questions to Make Data-Driven Decisions	November 2022 · Google
Computing in Python	February 2022 · Development Academy of the Philippines
Computing Microspecialization Pathway	September 2022 · Development Academy of the Philippines
Data Science and Machine Learning Using Python	September 2022 ·Development Academy of the Philippines
Data Visualization Microspecialization Pathway	September 2022 · Development Academy of the Philippines
Methods and Algorithms Microspecialization Pathway	September 2022 · Development Academy of the Philippines
Build Python Web Apps with Flask	August 2022 · DICT Philippines
Analyze Data with Python	July 2022 · DICT Philippines
Basic Statistics With Python	July 2022 · DICT Philippines
Experimental Design and Analysis	July 2022 · Development Academy of the Philippines
Programming for Beginners Using Python	July 2022 · Development Academy of the Philippines
Programming for Intermediate Users Using Python	July 2022 · DICT Philippines
Visualize Data with Python	July 2022 · DICT Philippines
Statistical Analysis and Modeling Using SQL and Python	May 2022 · Development Academy of the Philippines
Computing in Python	February 2022 · Development Academy of the Philippines
Data Visualization Using Tableau and Python	February 2022 · Development Academy of the Philippines
SQL for Business Users	February 2022 · Development Academy of the Philippines
Storytelling Using Data	December 2021 · Development Academy of the Philippines
Dashboards and Drill-Down Analytics	September 2021 · Development Academy of the Philippines
Data Visualization Fundamentals	September 2021 · Development Academy of the Philippines
Data Management Fundamentals	March 2021 · Development Academy of the Philippines
Essential Excel Skills for Data Preparation and Analysis	January 2021 · Development Academy of the Philippines
Getting Grounded on Analytics	December 2020 · Development Academy of the Philippines